CS 3710: Introduction to Cybersecurity (ICS) v1.0 Spring 2026

Instructor: Angela Orebaugh, angelao@virginia.edu

Class Meetings: MWF 11–11:50am Rice 130

Office Hours

Office hours posted on Piazza

TAs

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Course Overview

This course provides a comprehensive overview of essential cybersecurity concepts. It addresses both technical and non-technical concepts such as risks, threats, and vulnerabilities associated with the evolving cyber ecosystem. You will gain knowledge of fundamental cybersecurity principles and industry applications and examples. You will gain hands-on experience with realistic security scenarios and tools in a cloud-based lab environment.

Prerequisites

Must have completed CS 2150 or (CS 2100/CS2100 place out test **and** CS 2130) with grade of C- or higher.

Readings and Resource Requirements

You will be assigned relevant, current articles and papers accessible via the Internet or PDF. All course content is available on Canvas. You are not required to purchase a textbook but if you are looking for a good resource for cybersecurity foundational concepts you may use the following book:

Fundamentals of Information Systems Security 5th Edition, David Kim and Michael G. Solomon, 2026, ISBN: 9781284314977.

You will need a computer with a web browser and high speed Internet connection with the capability to run hands-on exercises in the Virginia Cyber Range.

Course Objectives

Students who complete this course will:

- Understand the ethical and policy context for cybersecurity in today's society
- Understand how to better safeguard one's personal computer
- Understand the basics of advanced topics in cybersecurity including encryption, digital forensics, binary exploits, and networks
- Understand the modern concepts in cybersecurity attacks and prevention

Use of AI

I am actively adapting this course for Responsible AI-Enhanced Learning, which includes Domain Knowledge (deep subject understanding), AI Knowhow (skillful use of AI tools), and Metacognition (self-awareness in learning). Each assignment will have information on AI use. Tools must be fully disclosed as a source along with the capacity in which they were used.

Assessment Components

You will be evaluated by lab assignments, programming assignments, cyber scenario analyses, Cyber Chat participation, and a final exam.

- **Lab Assignments (49 points)** – You will be responsible for completing 7 lab assignments (7 points each). These assignments help you develop critical thinking, problem solving, analysis, and hands-on experience. Evaluation Criteria – Your lab assignments will be evaluated on completeness, applying course concepts, interpreting results, and ability to achieve the desired outcome.
- **Programming Assignments (14 points)** – You will be responsible for completing 2 programming assignments (7 points each). These assignments help you develop critical thinking, problem solving, creativity, and programming skills. Evaluation Criteria – Your programming assignments will be evaluated on completeness, applying course concepts, applying critical thinking skills to create solutions to problems, and ability to achieve the desired outcome.
- **Cyber Scenario Analyses (CSA) (15 points)** – You will be responsible for completing 3 Cyber Scenario Analysis activities (5 points each). CSA's help assess your ability to evaluate real world cybersecurity scenarios and justify security decisions. Evaluation Criteria – CSA's will be evaluated using a provided rubric.
- **Cyber Chat (7 points)** – You will be responsible for presenting 1 Cyber Chat (7 points) lightning talk. You will sign up for a day to post a video of your presentation and associated writeup to the Piazza Cyber Chat folder. Select videos will be shown in class on Cyber Chat Fridays. You may choose any cybersecurity topic that interests you (i.e. a historical event, tool, exploit, recent breach, innovative research, etc.). This assignment will help you develop research, analysis, written communication, and verbal communication skills. Evaluation Criteria – Your Cyber Chat will be evaluated on comprehensiveness, analysis, and speaking and writing clarity and succinctness.
- **CTF Final Exam (15 points)** – You will be responsible for completing a final exam (15 points). The exam will be a capture the flag style exam using problem solving techniques and tools from the lab assignments and class lectures. The examination will be used to ensure the principles of the course have been mastered at an acceptable level. Evaluation Criteria – Final exams will be evaluated on your ability to solve problems and apply cybersecurity tools and techniques to identify the flag.

Assessment Component Honor Code and Late Policy

The UVA Honor Code is in effect for all assessment components and will automatically be included for assessment components submitted on Canvas. Academic integrity violations will not be tolerated. They include using someone else's Cyber Range account, copying from old assignments, working together with another student on assignments, and other violations. **Academic integrity violations will result in a 0 and possibly removal from class.**

Late submissions will receive a grade reduction of 10% per day. If you have any special circumstances, they need to be **communicated in advance**.

It is in the best interest of everyone in our community to keep the spread of infectious disease to a minimum. Students who are ill should stay home. This course has a very flexible learning environment including live streaming and recordings Please reach out to me if you have any concerns about assignments due to illness.

Assessment Component Allocation

Component	Points
Lab exercises	49 points (7 @7pts each)
Programming assignments	14 points (2 @7pts each)
CSAs	15 points (3 @5pts each)
Cyber Chat	7 points (1 @7pts each)
Final Exam	15 points (1 @15pts each)

Grading

A+	100	B	83–86	C-	70–72
A	95–99	B-	80–82	D+	67–69
A-	90–94	C+	77–79	D	63–66
B+	87–89	C	73–76	D-	60–62
				F	Below 60 (no credit)

Course Code of Ethics

All students in this course are expected to abide by ethical guidelines in their use of the information taught in the course. It is not the intent of the Department of Computer Science to teach any student how to create malicious code or how to exploit any software vulnerabilities. However, in order to teach students how to defend against malicious attacks upon computer systems, it is necessary to provide a certain amount of information about how such attacks are devised. Students are not to use any of this information to construct any such attacks themselves, or to render any computer system vulnerable to such attacks.

You are required to agree to abide by the following rules. If you cannot abide with these rules, you will have to drop this course.

1. You will not use the knowledge gained in this course to alter program or data or any other files belonging to another person without the express consent of that person.
2. You will not use the knowledge gained in this course to gain access to any computer system that you are not authorized, by the administrators of that system, to access.
3. You will not use the knowledge gained in this course to use any computer resources (e.g. processing time, disk space, network bandwidth, etc.) that you are not authorized, by the administrators of the affected systems, to use.

Course Schedule

Week	Date	Topic and Readings	Activities
Week 1	M–Jan 12	Course Overview and Introduction	
	W–Jan 13	Introduction to Virginia Cyber Range Password security Prep for Lab #1 John the Ripper	
	F–Jan 15	Internet of Things	Create account on Virginia Cyber Range Sign up for Cyber Chat
Week 2	M–Jan 19	No class – MLK Day	
	W–Jan 21	Networking, Recon, Scanning Prep for Lab #2 Nmap	
	F–Jan 23	Friday Cyber Chat	Lab #1 due Jan 23 @11:59pm
Week 3	M–Jan 26	Malicious Attacks, Threats, and Vulnerabilities	
	W–Jan 28	Malicious Attacks, Threats, and Vulnerabilities - continued	
	F–Jan 30	Malicious Attacks, Threats, and Vulnerabilities – continued	Lab #2 due Jan 30 @11:59pm
Week 4	M–Feb 2	Access Controls	
	W–Feb 4	Sniffing Prep for Lab #3 Wireshark	
	F–Feb 6	Friday Cyber Chat	
Week 5	M–Feb 9	CSA #1	
	W–Feb 11	Exploits Prep for Lab #4 Metasploit	
	F–Feb 13	Friday Cyber Chat	Lab #3 due Feb 13 @11:59pm
Week 6	M–Feb 16	Security Operations and Administration	
	W–Feb 18	Security Operations and Administration – continued Prep for Programming Assignment #1	
	F–Feb 20	Friday Cyber Chat	Lab #4 due Feb 20 @11:59pm
Week 7	M–Feb 23	Auditing, Testing, and Monitoring	
	W–Feb 25	Web Vulnerabilities Prep for Lab #5	
	F–Feb 27	Friday Cyber Chat	Programming Assignment #1 due Feb 27 @11:59pm
Week 8	M–Mar 2	Spring Break – no class	
	W–Mar 4	Spring Break – no class	
	F–Mar 6	Spring Break – no class	
Week 9	M–Mar 9	Cryptography	
	W–Mar 11	Cryptography – continued	
	F–Mar 13	Cryptography – continued	Lab #5 due Mar 13 @11:59pm
Week 10	M–Mar 16	Cryptography – continued	
	W–Mar 18	Prep for Lab #6 Mcrypt & GnuPG	
	F–Mar 20	Friday Cyber Chat	
Week 11	M–Mar 23	CSA #2	
	W–Mar 25	Network Security	

Week	Date	Topic and Readings	Activities
	F–Mar 27	Friday Cyber Chat	Lab #6 due Mar 27 @11:59pm
Week 12	M–Mar 30	Malicious Code and Activity	
	W–Apr 1	Malicious Code and Activity – continued	
	F–Apr 3	Binary Analysis Prep for Programming Assignment #2	
Week 13	M–Apr 6	Binary Analysis	
	W–Apr 8	Snort & iptables Prep for Lab #7 Snort iptables	
	F–Apr 10	Friday Cyber Chat	Programming Assignment #2 due Apr 10 @11:59pm
Week 14	M–Apr 13	CSA #3	
	W–Apr 15	Information Systems Security Education and Training Information Security Professional Certifications	
	F–Apr 17	Friday Cyber Chat	Lab #7 due Apr 17 @11:59pm
Week 15	M–Apr 20	Cybersecurity research	Final exam opens
	W–Apr 22	Cybersecurity research	
	F–Apr 24	Course wrap up	
Week 16	M–Apr 27		Final exam due Apr 27 by 11:59pm

Students with disabilities or learning needs

It is my goal to create a learning experience that is as accessible as possible. If you anticipate any issues related to the format, materials, or requirements of this course, please meet with me outside of class so we can explore potential options. Students with disabilities may also wish to work with the Student Disability Access Center (SDAC) to discuss a range of options to removing barriers in this course, including official accommodations. We are fortunate to have an SDAC advisor, Courtney MacMasters, physically located in Engineering. You may email her at cmacmasters@virginia.edu to schedule an appointment. For general questions please visit the SDAC website: *sdac.studenthealth.virginia.edu*. If you have already been approved for accommodations through SDAC, please send me your accommodation letter and meet with me so we can develop an implementation plan together

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Religious accommodations

It is the University's long-standing policy and practice to reasonably accommodate students so that they do not experience an adverse academic consequence when sincerely held religious beliefs or observances conflict with academic requirements.

Students who wish to request academic accommodation for a religious observance should submit their request to me by email as far in advance as possible. Students who have questions or concerns about academic accommodations for religious observance or religious beliefs may contact the University's Office for Equal Opportunity and Civil Rights (EOCR) at *UVAEOCR@virginia.edu* or 434-924-3200.

Harassment, Discrimination, and Interpersonal Violence

The University of Virginia is dedicated to providing a safe and equitable learning environment for all students. If you or someone you know has been affected by power-based personal violence, more information can be found on the UVA Sexual Violence website that describes reporting options and resources available - *www.virginia.edu/sexualviolence*.

The same resources and options for individuals who experience sexual misconduct are available for discrimination, harassment, and retaliation. UVA prohibits discrimination and harassment based on age, color, disability, family medical or genetic information, gender identity or expression, marital status, military status, national or ethnic origin, political affiliation, pregnancy (including childbirth and related conditions), race, religion, sex, sexual orientation, or veteran status. UVA policy also prohibits retaliation for reporting such behavior.

If you witness or are aware of someone who has experienced prohibited conduct, you are encouraged to submit a report to Just Report It (justreportit.virginia.edu) or contact EOCR, the office of Equal Opportunity and Civil Rights.

If you would prefer to disclose such conduct to a confidential resource where what you share is not reported to the University, you can turn to Counseling & Psychological Services ("CAPS") and Women's Center Counseling Staff and Confidential Advocates (for students of all genders).

As your professor and as a person, know that I care about you and your well-being and stand ready to provide support and resources as I can. As a faculty member, I am a responsible employee, which means that I am required by University policy and by federal law to report certain kinds of conduct that you report to me to the University's Title IX Coordinator. The Title IX Coordinator's job is to ensure that the reporting student receives the resources and support that they need, while also determining whether further action is necessary to ensure survivor safety and the safety of the University community.

Support for your career development

Engaging in your career development is an important part of your student experience. For example, presenting at a research conference, attending an interview for a job or internship, or participating in an extern/shadowing experience are not only necessary steps on your path but are also invaluable lessons in and of themselves. I wish to encourage and support you in activities related to your career development. To that end, please notify me by email as far in advance as possible to arrange for appropriate accommodations.

Student support team

You have many resources available to you when you experience academic or personal stresses. In addition to your professor, the School of Engineering and Applied Science has staff members located in Thornton Hall who you can contact to help manage academic or personal challenges. Please do not wait until the end of the semester to ask for help!

Learning

Lisa Lampe, Assistant Dean for Undergraduate Affairs
Georgina Nembhard, Director of Student Success
Courtney MacMasters, Accessibility Specialist
Free tutoring is available for most classes.

Health and Wellbeing

Kelly Garrett, Assistant Dean of Students, Student Safety and Support
Elizabeth Ramirez-Weaver, CAPS counselor*
Katie Fowler, CAPS counselor*

*You may schedule time with the CAPS counselors through Student Health (https://www.studenthealth.virginia.edu/getting-started-caps). When scheduling, be sure to specify that you are an Engineering student. You are also urged to use TimelyCare for either scheduled or on-demand 24/7 mental health care.

Community and Identity

The Center for Connection (The Connect) is a dedicated student space within UVA Engineering that fosters academic success and personal growth. Through its programs and initiatives, The Connect helps students strengthen their engineering identity while providing resources to help them thrive during their studies and beyond. Our work centers on three key areas: student belonging and development, academic support, and community programming grounded in intentional, data-driven strategies.

The Connect features an open study area, a flexible event space, and on-site staff who provide direct support and advising to students. It is part of the Office of Community, Opportunity, and Engagement.

Version change control:

1/7/26 v1.0 syllabus created.